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Inventor(s)	Kawada; Haruhiko et al.

Conveyance seat

Abstract

There is provided a conveyance seat that suppresses the deterioration of the sensitivity of a load sensor to an input load even when a heater wire and the load sensor are disposed to overlap each other. A conveyance seat includes a seat cushion that supports buttocks of an occupant; a seat back connected to the seat cushion to serve as a backrest for the occupant; a load sensor provided at the seat cushion or at a portion of the seat back; a heater wire disposed on an occupant side with respect to the load sensor; and an intermediate member that positions the heater wire. The intermediate member is disposed between the load sensor and the heater wire.

Inventors:	Kawada; Haruhiko (Tochigi, JP), Sasaki; Naohiro (Tochigi, JP), Tanabe; Yuma (Tochigi, JP), Ono; So (Tochigi, JP), Sonoda; Takamasa (Tochigi, JP), Shimazaki; Hayato (Tochigi, JP)
Applicant:	TS TECH CO., LTD. (Asaka, JP)
Family ID:	1000008821559
Assignee:	TS TECH CO., LTD. (Asaka, JP)
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USPC: 177/136

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7937789	12/2010	Feher	5/714	A47C 7/74
9333884	12/2015	Tachikawa	N/A	B60N 2/206
9550440	12/2016	Nagayasu	N/A	B60N 2/39
10556532	12/2019	Gallagher	N/A	B60N 2/5685
11225176	12/2021	Hayashi	N/A	B60N 2/4249
11697362	12/2022	Tanaka	297/188.1	B60N 2/3075
2002/0000742	12/2001	Wato et al.	N/A	N/A
2021/0323443	12/2020	Ozawa	N/A	B60N 2/5621

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201780174	12/2010	CN	N/A
2002-270338	12/2001	JP	N/A
WO 02/06083	12/2001	WO	N/A

OTHER PUBLICATIONS

Chinese Office Action (w/ English translation) for corresponding Application No. 202180021406.6, dated Mar. 6, 2024, 9 pages. cited by applicant

PCT International Search Report and Written Opinion (w/ English translation) for corresponding PCT Application No. PCT/JP2021/011749, mailed on Jun. 15, 2021, 10 pages. cited by applicant

Primary Examiner: Saint Surin; Jacques M

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage entry of International Application No. PCT/JP2021/011749 filed under the Patent Cooperation Treaty on Mar. 22, 2021, which claims priority to Japanese Patent Application No. 2020-051307 filed on Mar. 23, 2020, both of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a conveyance seat, particularly to a conveyance seat in which a seat back or a seat cushion is provided with a load sensor.

BACKGROUND ART

(3) There is a conveyance seat in which both a load sensor that detects that an occupant is seated and a heating element (heater) that heats the seat for heating are disposed.

(4) The load sensor provided in the conveyance seat is used to determine whether or not the occupant is seated in the conveyance seat, whether the occupant is an adult or a child, and in what posture the occupant is seated. The detection of the occupant by the load sensor is used in, for example, a seat belt reminder (SBR) system that urges the occupant to wear a seat belt when the seat belt is not worn.

(5) The load sensor is formed of a membrane switch including multiple contacts and, for example, is disposed between a trim cover (skin material) and a cushion pad forming a seat cushion of the conveyance seat.

(6) The heating element disposed inside the seat is a seat heater formed in a sheet shape and, for example, is disposed on the load sensor disposed on the cushion pad. The seat heater is configured such that a heater wire made of an electric resistor is disposed on a base material made of a material having a low thermal conductivity, for example, a non-woven fabric such as polyester resin. In a conveyance seat of the related art, the heater wire is directly disposed vertically above a contact portion (pressure sensitive portion) of the load sensor or in the vicinity of the contact portion, and even when a predetermined load is applied onto the contact portion of the load sensor, the seat heater does not bend and the sensitivity to detect a seating state of the occupant decreases, which is a problem.

(7) For this reason, for example, in a seat disclosed in Patent Literature 1, in consideration of an influence of the heater wire on the load sensor (referred to as a seating sensor in Patent Literature 1), the heater wire is disposed at a position where the heater wire does not overlap the load sensor.

CITATION LIST

Patent Literature

(8) PATENT LITERATURE 1: JP 2002-270338 A

SUMMARY OF INVENTION

Technical Problem

(9) However, when the heater wire is disposed not to overlap the load sensor, the deterioration of the sensitivity of the load sensor to an input load is unlikely to occur, but a disposition place of the heater wire is limited, so that the heating performance may be affected.

(10) Therefore, the present invention is conceived in view of the foregoing problems, and an object of the present invention is to provide a conveyance seat that suppresses the deterioration of the sensitivity of a load sensor to an input load even when a heater wire and the load sensor are disposed to overlap each other.

Solution to Problem

(11) In order to solve the foregoing problems, a conveyance seat according to the present invention includes: a seat cushion that supports buttocks of an occupant; a seat back connected to the seat cushion to serve as a backrest for the occupant; a load sensor provided at the seat cushion or at a portion of the seat back; a heater wire disposed on an occupant side with respect to the load sensor; and an intermediate member that positions the heater wire. The intermediate member is disposed between the load sensor and the heater wire.

(12) In the related art, an intermediate member that positions a heater wire used in a seat heater is disposed to be located on an occupant side with respect to the heater wire. For this reason, when the occupant is seated, the intermediate member is supported by the heater wire and is difficult to bend, and for this reason, even when a predetermined load is applied to a load sensor, a seating state may not be detected and the sensitivity may decrease.

(13) In the present invention, the intermediate member that positions the heater wire is disposed between the load sensor and the heater wire. For this reason, when the occupant is seated, the heater wire presses the intermediate member and the intermediate member bends, so that a switch of the load sensor is turned on. Since the heater wire is located on the occupant side with respect to the intermediate member, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed. For this reason, the heater wire can be disposed without considering an influence on the load sensor.

(14) In addition, in this configuration, the load sensor may include a contact portion that is turned on and off by an applied load, and the heater wire may be disposed at a position where the heater wire overlaps the contact portion of the load sensor when viewed from the occupant side.

(15) With this configuration, even when the heater wire overlaps the contact portion, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(16) In this configuration, a plurality of the heater wires may be disposed at positions where the heater wires overlap the contact portion.

(17) With this configuration, even when the plurality of heater wires overlap the contact portion of the load sensor, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(18) In this configuration, the heater wire may overlap the contact portion of the load sensor at a position offset from a center of the contact portion.

(19) With this configuration, even when the heater wire overlaps the contact portion of the load sensor at the position offset from the center of the contact portion, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(20) In this configuration, the heater wire may be disposed at a position where the heater wire overlaps a center of the contact portion of the load sensor.

(21) With this configuration, even when the heater wire is disposed at the position where the heater wire overlaps the center of the load sensor, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(22) In this configuration, the intermediate member may be a foundation fabric, and the heater wire may be fixed onto the foundation fabric.

(23) Since the intermediate member is the foundation fabric, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(24) In this configuration, the foundation fabric may be fixed to a surface of a pad forming the seat cushion or the seat back.

(25) Since the foundation fabric is fixed to the surface of the pad, the foundation fabric bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(26) In this configuration, the foundation fabric may be fixed to a back surface of a skin material forming the seat cushion or the seat back.

(27) Since the foundation fabric is fixed to the back surface of the skin material, the foundation fabric bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(28) In this configuration, the intermediate member may be a member having flexibility.

(29) Since the intermediate member has flexibility, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(30) In addition, in this configuration, the seat cushion and the seat back each may include a frame forming a skeleton, a pad that is a cushion material covering the frame, and a skin material covering the pad.

Advantageous Effects of Invention

(31) In the present invention, the intermediate member that positions the heater wire is disposed between the load sensor and the heater wire. For this reason, when the occupant is seated, the heater wire presses the intermediate member and the intermediate member bends, so that a switch of the load sensor is turned on. Since the heater wire is located on the occupant side with respect to the intermediate member, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed. For this reason, the heater wire can be disposed without considering an influence on the load sensor.

(32) In addition, with this configuration, even when the heater wire overlaps the contact portion, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(33) In addition, with this configuration, even when the plurality of heater wires overlap the contact portion of the load sensor, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(34) In addition, with this configuration, even when the heater wire overlaps the contact portion of the load sensor at the position offset from the center of the contact portion, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(35) In addition, with this configuration, even when the heater wire is disposed at the position where the heater wire overlaps the center of the load sensor, the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(36) In addition, since the intermediate member is the foundation fabric, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(37) In addition, since the foundation fabric is fixed to the surface of the pad, the foundation fabric bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(38) In addition, since the foundation fabric is fixed to the back surface of the skin material, the foundation fabric bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

(39) In addition, since the intermediate member has flexibility, the intermediate member bends easily, so that the deterioration of the sensitivity of the load sensor to an input load is suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a vehicle seat according to one embodiment of the present invention when viewed obliquely from front.

(2) FIG. 2 is a perspective view showing a frame of the vehicle seat according to one embodiment of the present invention.

(3) FIG. 3 is a view showing a seat cushion to which a load sensor and a seat heater are attached.

(4) FIG. 4A is a view showing a positional relationship between a contact portion of the load sensor and the seat heater and is a view showing a state where four heater wires overlap the contact portion.

(5) FIG. 4B is a view showing a positional relationship between the contact portion and the seat

heater and is a view showing a state where three heater wires overlap the contact portion.

(6) FIG. 4C is a view showing a positional relationship between the contact portion and the seat heater and is a view showing a state where a heater wire is disposed to overlap the contact portion at a position offset from a center of the contact portion.

(7) FIG. 4D is a view showing a positional relationship between the contact portion and the seat heater and is a view showing a state where a heater wire is disposed at a position where the heater wire overlaps the center of the contact portion.

(8) FIG. 5A is a cross-sectional view taken along line V-V of FIG. 3 and is a view showing a positional relationship between the contact portion of the load sensor and the seat heater.

(9) FIG. 5B is a cross-sectional view showing a state where a load is applied to the contact portion of the load sensor and the contact portion is turned on.

(10) FIG. 6 is a cross-sectional view showing a positional relationship between a seat heater and a load sensor in the related art.

(11) FIG. 7 is a graph that compares ON loads of the load sensors.

(12) FIG. 8 is a perspective view showing an example in which the load sensor and the seat heater are attached to a seat back.

DESCRIPTION OF EMBODIMENTS

(13) Hereinafter, a configuration of a conveyance seat according to one embodiment of the present invention (present embodiment) will be described with reference to the drawings. Meanwhile, the embodiment to be described below is one example for facilitating understanding of the present invention and does not limit the present invention. Namely, the present invention can be changed or improved without departing from the concept thereof, and it goes without saying that the present invention includes equivalents thereof.

(14) Incidentally, hereinafter, a vehicle seat will be provided as one example of the conveyance seat, and a configuration example of the vehicle seat will be described. Meanwhile, the present invention can also be applied to a conveyance seat other than the vehicle seat, for example, to a seat mounted in a ship or an aircraft.

(15) In addition, in the following description, a “front to back direction” is a front to back direction of the vehicle seat and is a direction coinciding with a traveling direction of a vehicle in traveling. In addition, a “seat width direction” is a width direction of the vehicle seat and is a direction coinciding with a right to left direction when viewed from an occupant seated in the vehicle seat. In addition, an “up to down direction” is an up to down direction of the vehicle seat and is a direction coinciding with a vertical direction when the vehicle travels on a horizontal surface.

(16) In addition, in the following description, when various directions such as the “seat width direction” and a “seat height direction” are described with “seat” added thereto, the various directions indicate directions with respect to the vehicle seat, and when directions are described with “vehicle” such as “vehicle inside” and “vehicle outside” added thereto, the directions indicate directions with respect to the vehicle.

(17) In addition, an “occupant side” indicates a side where the occupant stays when the occupant is seated in a state where the occupant can be seated on the vehicle seat, and means a front side of a seat back in the seat back of the vehicle seat and means an upper side of a seat cushion in the seat cushion.

(18) In addition, unless otherwise specified, the shape, the position, the posture, and the like of each portion of the vehicle seat to be described below will be described based on a case where the occupant can be seated in the vehicle seat.

(19) <<Basic Configuration of Vehicle Seat According to Present Embodiment>>

(20) A configuration of a vehicle seat according to the present embodiment (hereinafter, a vehicle seat S) will be described with reference to FIGS. 1 to 3. FIG. 1 is a perspective view of the vehicle seat S when viewed obliquely from front, FIG. 2 is a perspective view showing a seat frame F of the vehicle seat S, and FIG. 3 is a plan view showing a seat cushion S1 of the vehicle seat S to

which a load sensor **3** and a seat heater **4** are attached. In FIG. **1**, for convenience of illustration, a portion of the vehicle seat **S** is illustrated in a state where trim covers **14** and **24** that are skin members are removed. In addition, FIG. **3** shows the seat cushion **S1** with the trim cover **14** removed.

(21) The vehicle seat **S** is a seat which is placed on a vehicle body floor and in which the occupant of the vehicle is seated. In the present embodiment, the vehicle seat **S** is used as a front seat corresponding to a front seat of the vehicle. However, the vehicle seat **S** is not limited to the front seat, and the vehicle seat **S** may be a rear seat of the vehicle and can also be used as a middle seat of a second row or a rear seat of a third row of a vehicle in which three rows of seats are provided in the front to back direction.

(22) As shown in FIG. **1**, the vehicle seat **S** includes, as main components, the seat cushion **S1** serving as a seating portion that supports the buttocks of the occupant, a seat back **S2** serving as a backrest that supports the back of the occupant, and a headrest **S3** that supports the head of the occupant. The seat cushion **S1** and the seat back **S2** are connected to each other with a reclining mechanism **10** (refer to FIG. **2**) interposed therebetween. In addition, a slide rail **R** is installed below the seat cushion **S1**, and the seat cushion **S1** is placed on the vehicle body floor in a state where the vehicle seat **S** is slidable in the front to back direction by the slide rail **R** (refer to FIG. **2**).

(23) The seat frame **F** shown in FIG. **2** is provided inside the vehicle seat **S**, and the seat frame **F** mainly includes a cushion frame **1** and a back frame **2**. The cushion frame **1** forms a skeleton of the seat cushion **S1** in the seat frame **F**, and the back frame **2** forms a skeleton of the seat back **S2** in the seat frame **F**. A rear end portion of the cushion frame **1** is connected to a lower end portion of the back frame **2** through the reclining mechanism **10**.

(24) The seat cushion **S1** is formed by placing a cushion pad **13** that is a cushion material on the cushion frame **1** and by covering the cushion pad **13** with the trim cover **14**. The seat back **S2** is formed by placing a back pad **23** that is a cushion material on the back frame **2** and by covering the back pad **23** with the trim cover **24**.

(25) The cushion pad **13** and the back pad **23** are, for example, a urethane base material molded by foam molding using a urethane foaming agent. In addition, the trim covers **14** and **24** are made of fabric, film, cloth, leather, sheet, or the like and are attached to cover the cushion pad **13** and the back pad **23**, respectively, in a state where the trim covers **14** and **24** are stretched to be subjected to a predetermined tension.

(26) In the present embodiment, as shown in FIG. **2**, the cushion frame **1** has a substantially square frame shape as an outer shape. Then, the cushion frame **1** includes, as main components, a pair of cushion side frames **12** forming respective right and left end portions in the seat width direction, a pan frame **11** forming a front end portion of the cushion frame **1**, a connection pipe **15** connecting the right and left cushion side frames **12** at a rear end portion, and a spring **16**.

(27) The back frame **2** is mainly made of a pipe processed into a square frame shape, and as shown in FIG. **2**, the back frame **2** includes an upper frame **21** having an inverted U shape, a pair of back side frames **22** forming right and left end portions in the seat width direction, and a lower frame **26** connecting lower end portions of the pair of back side frames **22**. In addition, a connection frame **28** that connects portions of the upper frame **21** is provided, the portions extending up and down. Two headrest guides **29** are provided at a center of an upper end of the upper frame **21**, and a headrest pillar **27** extending from the headrest **S3** can be inserted into and held by the two headrest guides **29**.

(28) An exemplary example of a forming material of each member forming the cushion frame **1** and the back frame **2** is a material having a sufficient rigidity such that each member is not greatly deformed when receiving a load, for example, a metal material such as steel or an aluminum alloy. Means for joining the members forming the cushion frame **1** and the back frame **2** is welding, but as the joining means, joining with bolts and an adhesive agent may be used in combination.

(29) The seat heater **4** that is a heating element is provided between the cushion pad **13** and the trim

cover **14** of the seat cushion **S1**. In addition, the load sensor **3** is disposed on a lower side (side opposite the occupant side) of the seat heater **4**. Hereinafter, the load sensor **3** and the seat heater **4** provided in the vehicle seat **S** of the present embodiment will be described with reference to FIGS. **3** to **5B**.

(30) <<Load Sensor>>

(31) The load sensor **3** is a sensor that detects that the occupant is seated on the seat cushion **S1** of the vehicle seat **S**. The load sensor **3** is a membrane switch and, as shown in FIG. **3**, includes a plurality of contact portions **31**, a communication portion **32** communicating with the contact portions **31**, a circuit unit **33**, a harness **34**, and an inlet and outlet port **35**. As shown in FIG. **5A**, each of the contact portions **31** is configured such that a pair of electrodes **31C** and **31D** are disposed at a predetermined interval inside surface materials **31A** and **31B** made of a pair of films, to face each other. Namely, an air layer **31E** is formed between the pair of electrodes **31C** and **31D**. An adhesive material **31F** including a resin spacer is disposed at a portion at which the electrodes **31C** and **31D** do not exist. In a state where a load is not applied to the contact portion **31**, the interval between the electrodes **31C** and **31D** is kept by the spacer of the adhesive material **31F**. Then, the electrodes **31C** and **31D** disposed up and down are configured to come into contact with each other when a load is applied to the contact portion **31** from above (refer to FIG. **5B**). In other words, the contact portion **31** can be turned on and off by the applied load, and a switch of the contact portion **31** is turned on and off depending on the magnitude of the applied load.

(32) Each of the contact portions **31** is connected to the inlet and outlet port **35** through the communication portion **32** that is an air passage, and air between the electrodes **31C** and **31D** located on upper and lower sides of the contact portion **31** comes in and out through the communication portion **32** and the inlet and outlet port **35**.

(33) The circuit unit **33** is connected to the contact portions **31** to detect an ON signal indicating that the electrodes **31C** and **31D** on the upper and lower sides of each of the contact portions **31** have come into contact with each other, and to output the detected ON signal to an electronic control unit (ECU) attached to the vehicle through the harness **34** and through the circuit unit **33** to which an end portion of the harness **34** is connected. In a case where the ECU has received an ON signal, for example, when another sensor detects that a seat belt is not fastened, the occupant is urged to fasten the seat belt.

(34) <<Seat Heater>>

(35) As shown in FIGS. **3**, **5A**, and **5B**, the seat heater **4** is disposed above the contact portion **31** of the load sensor **3**. The seat heater **4** includes a foundation fabric **41** (one example of an intermediate member) having a sheet shape, a heater wire **42**, a circuit unit **43**, and a harness **44**. The foundation fabric **41** is made of a material having a low thermal conductivity, for example, a non-woven fabric such as polyester resin. In addition, the foundation fabric **41** has flexibility and is configured to bend when being pressed from above (refer to FIG. **5B**). In the present embodiment, the foundation fabric **41** is used as an intermediate member that positions the heater wire **42**, but the intermediate member is not limited to the foundation fabric **41** and may be, for example, a rubber sheet having elasticity.

(36) The heater wire **42** is disposed in a meandering shape to cover the entire region of the foundation fabric **41** as shown in FIG. **3**, and the position of the heater wire **42** is determined by being fixed onto the foundation fabric **41** with an adhesive agent or the like. The seat heater **4** is formed of a radiant heater formed of the foundation fabric **41** and the heater wire **42**. The heater wire **42** is formed of, for example, an electric resistor made of copper, a copper alloy, or the like. The circuit unit **43** causes an electric current to flow to the heater wire **42** through the harness **44**, so that heat is generated in the heater wire **42**.

(37) In the vehicle seat **S** of the present embodiment, as shown in FIG. **5A**, the load sensor **3** is disposed on the cushion pad **13** of the seat cushion **S1**. Then, the foundation fabric **41** of the seat heater **4** is disposed on the load sensor **3**, and the heater wire **42** is disposed on the foundation

fabric **41**. The trim cover **14** as a skin material is attached onto the seat heater **4**. In other words, the heater wire **42** of the seat heater **4** is disposed on a side where the occupant is located with respect to the load sensor **3** (occupant side), and the foundation fabric **41** is disposed between the heater wire **42** and the load sensor **3**.

(38) The foundation fabric **41** is fixed to the cushion pad **13** by an adhesive agent or sewing. The foundation fabric **41** may be fixed to the trim cover **14** that is a skin material.

(39) Since the seat heater **4** is disposed on the load sensor **3**, as shown in FIG. 3, the heater wire **42** is disposed to overlap the contact portion **31** of the load sensor **3**. The heater wire **42** meanders and an interval L between the heater wires **42** adjacent to each other varies depending on the location. For this reason, when the interval between the heater wires **42** is smaller than the size (diameter D) of the contact portion **31**, for example, as shown in FIG. 4A, four heater wires **42** may be disposed to overlap the contact portion **31**, namely, four heater wires may be disposed to cross over the contact portion **31**. As shown in FIG. 4B, three heater wires may cross over the contact portion **31**. As shown in FIG. 4C, one heater wire **42** may be disposed to cross over the contact portion **31** at a position offset from a center C of the contact portion **31**. As shown in FIG. 4D, the heater wire **42** may be disposed at a position where the heater wire **42** overlaps the center C of the contact portion **31**.

(40) In the vehicle seat S of the present invention, when the seat heater **4** is disposed on the contact portion **31** of the load sensor **3**, as shown in FIG. 5A, the foundation fabric **41** is disposed on the electrodes **31C** and **31D** interposing the air layer **31E** of the contact portion **31** therebetween, and the heater wires **42** are disposed on the foundation fabric **41**.

(41) For this reason, when the occupant is seated to apply a load downward (in a direction of arrow A), the load is concentrated on the heater wires **42**, thereby pressing the foundation fabric **41** downward as shown in FIG. 5B, and thereby bringing the electrodes **31C** and **31D** of the contact portion **31** of the load sensor **3** into contact with each other to turn on the switch.

(42) As shown in FIG. 6, a vehicle seat of the related art has a configuration in which heater wires **142** are disposed on a load sensor **103** and a foundation fabric **141** is disposed on the heater wires, namely, on the occupant side. For this reason, when the occupant is seated to apply a load downward (in a direction of arrow B of FIG. 6), the load is distributed by the foundation fabric **141**, so that the heater wires **42** serve as beams that support the foundation fabric **141** so as not to bend. For this reason, in a case where the foundation fabric **141** is located above the heater wires **42**, even when a predetermined load is applied, the foundation fabric **141** does not bend and, in a contact portion **131** of the load sensor **103**, electrodes **131C** and **131D** interposed between surface materials **131A** and **131B** do not come into contact with each other, so that a switch of the contact portion **131** may be difficult to turn on.

(43) For this reason, a configuration is employed in which as shown in FIG. 5A, the foundation fabric **41** is pressed via the heater wires **42** by disposing the foundation fabric **41** between the load sensor **3** and the heater wires **42**, so that the foundation fabric **41** bends and the switch of the contact portion **31** of the load sensor **3** is easily turned on.

(44) FIG. 7 shows a graph that compares a load required to turn on the load sensor **103** when the foundation fabric **141** is disposed on the heater wires **142** in the related art (hereinafter, referred to as an ON load) and an ON load when the heater wires **42** are disposed on the foundation fabric **41** used in the vehicle seat S of the present embodiment. The case where the foundation fabric **141** is disposed on the heater wires **142** in the related art is shown by a dotted line, and the case where the foundation fabric **41** is disposed below the heater wires **42** in the present embodiment is shown by a solid line.

(45) As shown in FIG. 7, it can be confirmed that when four heater wires **42** are disposed on the contact portion **31** and when three heater wires **42** are disposed on the contact portion **31**, the ON load is greatly reduced, and the sensitivity of the load sensor **3** to an input load is improved by disposing the foundation fabric **41** below the heater wires **42**. Therefore, even when the heater

wires **42** are disposed to overlap the load sensor **3**, the deterioration of the sensitivity of the load sensor **3** to an input load is suppressed. For this reason, the heater wires **42** do not need to be disposed to avoid the load sensor **3**, and the heater wires **42** can be disposed without considering an influence on the load sensor **3**.

(46) In addition, it can be confirmed that even when one heater wire **42** crosses over the contact portion **31**, the width of reduction in ON load is small, but the ON load is reduced similarly to when four or three heater wires **42** cross over the contact portion **31**, and the sensitivity of the load sensor **3** to an input load is improved by disposing the foundation fabric **41** below the heater wires **42**. The ON load is reduced and the sensitivity to an input load is improved by disposing the heater wires **42** on the foundation fabric **41**, so that the detection performance of the load sensor **3** is improved and the switch is easily turned on.

(47) FIG. **8** shows another example of the vehicle seat **S**. In the vehicle seat **S** shown in FIG. **1**, the load sensor **3** and the seat heater **4** are disposed on the seat cushion **S1**, but as shown in FIG. **8**, the load sensor **3** and the seat heater **4** may be disposed on the seat back **S2**. At this time, the load sensor **3** is used as, for example, a sensor of a system that measures a heart rate or a respiratory rate of the occupant.

(48) The load sensor **3** and the seat heater **4** are disposed on a front surface side of the seat back **S2**, namely, on the occupant side. The load sensor **3** is disposed on the front surface side of the back pad **23** of the seat back **S2**, and the heater wires **42** of the seat heater **4** are disposed on the occupant side with respect to the load sensor **3**. The foundation fabric **41** is disposed between the load sensor **3** and the heater wires **42**. Since the heater wires **42** are disposed on the occupant side with respect to the foundation fabric **41**, the ON load of the load sensor **3** is reduced and the sensitivity of the load sensor **3** to an input load is improved as compared to when the heater wires **42** are disposed on a rear side with respect to the foundation fabric **41**. For this reason, the detection performance of the load sensor **3** is improved, and when the occupant is seated, the load sensor **3** is easily turned on.

(49) In addition, the foundation fabric **41** is fixed to the back pad **23** of the seat back **S2** using an adhesive agent or the like. The foundation fabric **41** may be fixed to the trim cover **24** (skin material) attached to cover the back pad **23**.

REFERENCE SIGNS LIST

(50) **S**: vehicle seat (conveyance seat) **S1**: seat cushion **S2**: seat back **S3**: headrest **F**: seat frame **1**: cushion frame **10**: reclining mechanism **11**: pan frame **12**: cushion side frame **13**: cushion pad **14**: trim cover (skin material) **15**: connection pipe **16**: spring **2**: back frame **21**: upper frame **22**: back side frame **23**: back pad **24**: trim cover (skin material) **27**: headrest pillar **28**: connection frame **29**: headrest guide **3**, **103**: load sensor **31**, **131**: contact portion **31A**, **131A**: surface material **31B**, **131B**: surface material **31C**, **131C**: electrode **31D**, **131D**: electrode **31E**: air layer **31F**: adhesive material **C**: center of contact portion **32**: communication portion **33**: circuit unit **34**: harness **35**: inlet and outlet port **4**: seat heater **41**, **141**: foundation fabric (intermediate member) **42**, **142**: heater wire **43**: circuit unit **44**: harness

Claims

1. A conveyance seat, comprising: a seat cushion that supports buttocks of an occupant; a seat back connected to the seat cushion to serve as a backrest for the occupant; a load sensor provided at the seat cushion or at a portion of the seat back; a plurality of heater wire, wherein a heater wire is disposed on an occupant side with respect to the load sensor; and an intermediate member that positions the heater wire, wherein: the seat cushion and the seat back each include a pad and a skin material covering the pad, the load sensor includes a contact portion that is turned on and off by an applied load, the intermediate member is disposed between the load sensor and the heater wire, the heater wire is in direct contact with the back surface of the skin material, the plurality of the heater

wires is disposed at positions where the heater wire overlaps the contact portion of the load sensor when viewed from the occupant side.

2. The conveyance seat according to claim 1, wherein the heater wire overlaps the contact portion of the load sensor at a position offset from a center of the contact portion.

3. The conveyance seat according to claim 1, wherein the heater wire is disposed at a position where the heater wire overlaps a center of the contact portion of the load sensor.

4. The conveyance seat according to claim 1, wherein the intermediate member is a foundation fabric, and the heater wire is fixed onto the foundation fabric.

5. The conveyance seat according to claim 4, wherein the foundation fabric is fixed to a surface of a pad forming the seat cushion or the seat back.

6. The conveyance seat according to claim 4, wherein the foundation fabric is fixed to a back surface of a skin material forming the seat cushion or the seat back.

7. The conveyance seat according to claim 1, wherein the intermediate member is a member having flexibility.

8. The conveyance seat according to claim 1, wherein the seat cushion and the seat back each include a frame forming a skeleton, a pad that is a cushion material covering the frame, and a skin material covering the pad and the frame is covered by the pad.

9. A method of manufacturing a conveyance seat, comprising: preparing a seat cushion that supports buttocks of an occupant, a seat back that serves as a backrest for the occupant, a load sensor includes a contact portion that is turned on and off by an applied load, a heater wire, and an intermediate member that positions the heater wire, connecting the seat back to the seat cushion, providing the load sensor at the seat cushion or at a portion of the seat back, and disposing the intermediate member between the load sensor and the heat wire, wherein the seat cushion and the seat back each include a pad and a skin material covering the pad, the heater wire is in direct contact with the back surface of the skin material, and a plurality of the heater wires are disposed at positions where the heater wire overlaps the contact portion of the load sensor when viewed from the occupant side.
